The First EgoCross Challenge at EgoVis 2026: Cross-Domain Egocentric Video Question Answering

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Abstract

EgoCross is a cross-domain egocentric video question answering benchmark designed to evaluate whether multimodal large language models can generalize beyond common daily-life scenarios. The first EgoCross Challenge was hosted at the Third EgoVis Workshop at CVPR 2026 and evaluated models on first-person videos from four target domains: surgery, industrial assembly, extreme sports, and animal perspectives. Each test example consists of an egocentric video clip, a question, and four candidate answers, from which the model must select the correct option. This technical report introduces the challenge task, benchmark resources, and two official Codabench tracks. The Source-Limited Track restricts participants to the official baseline model and a small support set, whereas the Open-Source Track permits broader choices of models and training data under rules that prohibit the manual construction of target-domain training data. In total, the challenge received more than 1,500 submissions from over 130 participants, with 19 teams participating in the Open-Source Track and 38 teams in the Source-Limited Track. We further present the official leaderboard results and summarize the winning solutions from both tracks. We hope that this report will serve as a useful technical reference for advancing cross-domain egocentric video understanding. All resources, including the challenge data, baseline implementation, and code released by the winning teams, are made publicly available.

1. Introduction

Egocentric video understanding [1–9] aims to interpret the world from a first-person perspective. By capturing visual experiences from the viewpoint of the camera wearer, egocentric videos provide rich cues about human attention, intention, interactions, and task progression. These distinctive characteristics make egocentric video understanding fundamental to a wide range of applications, including wearable assistants [10–12], embodied AI [6, 13–15], navigation [16], robotics [17–25], and augmented reality [26–28].

However, most existing egocentric benchmarks [8, 29–35, 35, 36] primarily focus on common daily activities, such as cooking, household work, and routine object interactions. Although these settings provide an important foundation, real-world first-person applications often involve specialized domains with distinct visual appearance, terminology, temporal structure, and operational risks. A model that recognizes everyday objects may still fail to identify surgical instruments, understand circuit-board assembly, anticipate skiing motions, or interpret animal-mounted footage.

To investigate how models cope with these challenging yet practically important domain gaps, EgoCross [37] extends domain adaptation research from conventional vision tasks, such as image classification [38–48], object detection [49–53], and semantic segmentation [54–59], to the more advanced setting of egocentric video question answering. Specifically, EgoCross introduces a cross-domain egocentric VQA benchmark spanning four target domains: surgery, industry, extreme sports, and animal perspectives. The benchmark contains 798 video clips and 957 question–answer pairs, providing, for the first time, a comprehensive testbed for evaluating egocentric VQA under substantial domain shifts.

As EgoCross was only recently introduced, the first

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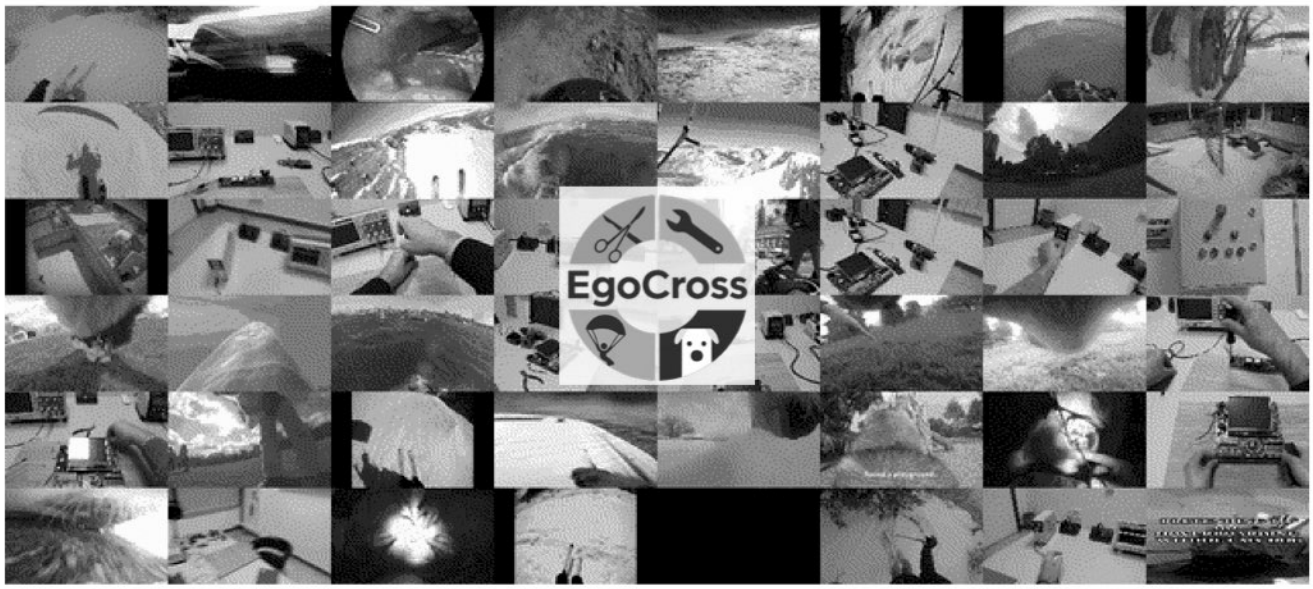


Figure 1. **EgoCross task examples.** The challenge covers professional and non-daily egocentric domains, including surgery, industry, extreme sports, and animal perspectives, with each question categorized by its corresponding task type.

EgoCross Challenge[1] was formally organized as part of the Third ***EgoVis Workshop*** at CVPR 2026[2] to increase its visibility within the community and stimulate further research on cross-domain egocentric VQA. EgoVis brings together representative research, benchmarks, and challenges from the egocentric vision community, making it a natural venue for assessing progress in cross-domain first-person video understanding. The challenge comprises two official Codabench competitions: the *Source-Limited Track* and the *Open-Source Track*. These tracks are designed to investigate complementary research questions. The Source-Limited Track emphasizes controlled and fair comparisons of adaptation methods under a fixed baseline model and a small target-domain support set. In contrast, the Open-Source Track evaluates how far stronger architectures, broader pretraining resources, and additional permissible data can advance cross-domain egocentric video question answering.

Overall, the challenge received more than 1,500 submissions from over 130 participants, with 19 teams participating in the Open-Source Track and 38 teams in the Source-Limited Track. This technical report provides a comprehensive overview of the first EgoCross Challenge. Section 2 describes the challenge design, including the task formulation, competition tracks, dataset composition, baseline model, and evaluation protocol. Section 3 reports and analyzes the official leaderboard results. Section 4 summarizes the winning solutions proposed by participating teams in both the Open-Source and Source-Limited Tracks.

[1]EgoCross: https://egocross-benchmark.github.io/

[2]EgoVis: https://egovis.github.io/cvpr26/

2. EgoCross Challenge

2.1. Challenge Overview

Building on the EgoCross benchmark introduced above, the challenge aims to evaluate and promote cross-domain egocentric video understanding. We adopt a closed-ended video question answering setting, referred to as *CloseQA*, in which each example consists of an egocentric video clip, a natural-language question, and four candidate answers. Participants are required to select the correct answer for each example. Note that the original EgoCross benchmark [37] provides both closed-ended and open-ended question answering formats for each question. However, considering the computational cost of evaluation and the reliability of automatic metrics, the challenge focuses exclusively on the CloseQA setting.

The challenge includes two official Codabench tracks: the *Source-Limited Track*[3], which restricts participants to the official baseline model and provides a small, fixed support set for model adaptation; and the *Open-Source Track*[4], which allows broader choices of models and training data while prohibiting the manual construction of target-domain training data. Fig. 2 illustrates the key differences between the two tracks.

2.2. Task Formulation

Given an egocentric video clip V, a question q, and four candidate answers $\mathcal{A} = \{a_A, a_B, a_C, a_D\}$, a model predicts

[3]Source-Limited Track: https://www.codabench.org/competitions/11279/

[4]Open-Source Track: https://www.codabench.org/competitions/13868/